**SNJB's**

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**College of Engineering**

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ESTD - 1928

# DBMSL Mini Project

**Course Code : 24-PCC-CS-2-07**

**Course Name : Database Management System Lab**

**Second Year Bachelor of Technology**

**A.Y. :- 20 - 20**

| | |
|---|---|
| **Name/Names of the Student** | |
| **Department / Program** | |
| **Division and Batch** | |
| **Roll No/Nos** | |
| **Semester** | |
| **Course Incharge Name** | |
| **GitHub Upload Link** | |

**Department of Computer Engineering**

**Rubrics for Group C Project**

| Criteria | Excellent (10-9) | Very Good (8-7) | Good (6-5) | Satisfactory (4-3) | Poor (2-1) |
|---|---|---|---|---|---|
| **Understanding of Front-End Technology** | Demonstrates deep understanding; UI well designed; correct integration. | Good understanding; UI clear and functional. | Basic UI; minor design/logic issues. | Minimal UI; partially functional. | No clear understanding; UI not functional. |
| **Database Connectivity (MongoDB /MySQL)** | Connection implemented efficiently with no errors; follows best practices. | Connection works with minimal issues. | Connectivity implemented with minor flaws. | Partially working or contains errors. | Connectivity missing or completely incorrect. |
| **Implementation of CRUD Operations** | All CRUD operations correctly implemented and tested. | Most CRUD operations implemented correctly. | Some CRUD operations implemented. | Only one or two operations implemented. | CRUD operations missing or incorrect. |
| **Documentation / Report Preparation and Github Upload** | Well-structured, clear, and complete manual with theory, aim, code, and conclusion. Code Uploaded on GitHub | Good documentation with minor omissions. Code Uploaded on GitHub | Sufficient documentation; lacks clarity. Code Uploaded on GitHub | Incomplete documentation .Code not Uploaded on GitHub | Very poor or missing documentation. Code not Uploaded on GitHub |

## 1.Title Of The Project

Project Title: Fitness Application

Technology Used (Frontend + Backend):

- Frontend: Java (Swing / Console-based UI)
- Backend: Java using Java Database Connectivity
- Database: MySQL

## 2.Abstract:-

The Fitness Application is designed to help users manage their daily fitness activities, including workouts, diet plans, and health progress tracking. In today's fast-paced lifestyle, maintaining fitness records manually is difficult and inefficient. This system provides a digital solution to store and manage user fitness data efficiently.

With increasing awareness about health and fitness, people need tools to track their physical activities. Manual tracking is error-prone and inefficient, hence a digital system is required.

- **Problem statement**

In today's busy lifestyle, individuals often struggle to maintain and track their fitness activities such as workouts, diet, and health progress. Traditional methods of recording fitness data manually are inefficient, time-consuming, and prone to errors. There is a need for a digital system that can efficiently store, manage, and retrieve fitness-related data in an organized manner.

- **Objective**
- To develop a fitness application that helps users manage their workout and diet records
- To provide easy options to add, update, delete, and view fitness data
- To store user information securely using a database
- To improve accuracy and reduce manual effort in tracking fitness progress

- **Technologies used**

- Frontend: Java (Swing / Console-based interface)
- Backend: Java using Java Database Connectivity
- Database: MySQL
- Tools/IDE: Eclipse / IntelliJ IDEA / NetBeans
- Database Tool: MySQL Workbench

- **Expected outcome**
- A fully functional fitness management system
- Efficient storage and retrieval of user fitness data
- Improved tracking of workouts, diet, and progress
- Reduction in manual errors and data redundancy
- A scalable system that can be enhanced into a web or mobile application

# 3 Introduction

**Background of the project**
**Need of the system**

- Maintain fitness records digitally
- Easy tracking of workouts and diet
- Reduce manual errors
- Quick access to data
- **Real-life application**
- Gym management systems
- Personal fitness tracking apps
- Health monitoring systems
- **Scope of the project**
- Manage user fitness data
- Track workouts and diet
- Store and retrieve data efficiently
- Can be extended to mobile/web applications

# 4.Softare Requirement Specification

## 4.1 Functional Requirements

- Add new fitness record
- Update existing record
- Delete record
- Search and view user data

## 4.2 Non-Functional Requirements

**Performance**

- Fast data retrieval and storage
- Efficient database operations

**Security**

- Secure database connection
- Use of PreparedStatement to prevent SQL injection

**Usability**

- Simple and user-friendly interface
- Easy navigation

## 4.3 Hardware & Software Requirements

**Hardware:**

- Minimum 4 GB RAM
- Processor: Intel i3 or higher **OS:**
- Windows / Linux / macOS **Frontend:**
- Java Swing / Console **Backend:**
- Java (JDBC) **Tools/IDE:**
- Eclipse / IntelliJ IDEA / NetBeans
- MySQL Workbench

## 5. Normalization

**1NF (First Normal Form):**

- No repeating groups
- Atomic values

**2NF (Second Normal Form):**

- No partial dependency
- All attributes depend on primary key

**3NF (Third Normal Form):**

- No transitive dependency
- Data is fully normalized

## Conclusion Summary Of The Project

The Fitness Application successfully provides a system for managing fitness-related data using Java and MySQL. It simplifies the process of tracking workouts, diet, and health progress.

**Future Scope**

- Mobile application (Android/iOS)
- Integration with wearable devices
- Cloud database support
- AI-based fitness recommendations